

Sri Lanka Institute of Information Technology



Bank Marketing Analysis Final Report

IT3101 - Data Warehousing and Business Intelligence

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1. Introduction

1.1 Business domain and operational context

The selected business domain is Retail Banking and Financial Services, focusing specifically on Direct Telemarketing Campaigns for Term Deposit Subscriptions.

The selected Retail Banking Direct Marketing campaign dataset represents a real-world operational environment well-suited for a dedicated Data Warehouse and Business Intelligence implementations. Direct telemarketing for banking products operates across fragmented transaction points such as operational CRM platforms, telephony call logging systems, and external macroeconomic reporting indices.

In retail banking, personal term deposits represent a vital mechanism for capital acquisition and liquidity management. Historically, financial institutions have relied on mass outbound telemarketing to pitch deposit products. However, indiscriminate cold-calling leads to customer churn, high operational call-center expenditure, and poor conversion efficiency.

1.2 Operational business problem

The operational institution conducts large-scale outbound phone campaigns. We have identified several critical bottlenecks:

- Inability to evaluate agent efficacy and call duration ROI in real time.
- Lack of demographic segmentation, causing agents to target low-propensity prospects.
- Fragmented operational silos where customer demographic data, call center interaction logs, and external macroeconomic indicators reside in disconnected systems.
- Inefficient follow-up scheduling, where the timing between campaigns (pdays) is unmonitored.

The business objective is to centralize disparate operational logs into an Enterprise Data Warehouse and deploy Business Intelligence dashboards to identify high-propensity customer segments, monitor historical conversion rates, and optimize outbound marketing schedules.

1.3 Dataset source, Formats and volume

The data foundation represents an operational banking environment containing 41,188 historical contact records collected across multi-channel outreach campaigns.

- **Source Systems & Formats:** 4 operational feeds consisting of CSV files, converted JSON operational application logs, and external reference tables.
- **Primary Entity Scope:** 41,188 individual customer outreach records (Customer IDs 1 to 41,188).
- **Target Feature:** Term deposit subscription status mapped downstream to a binary analytical metric

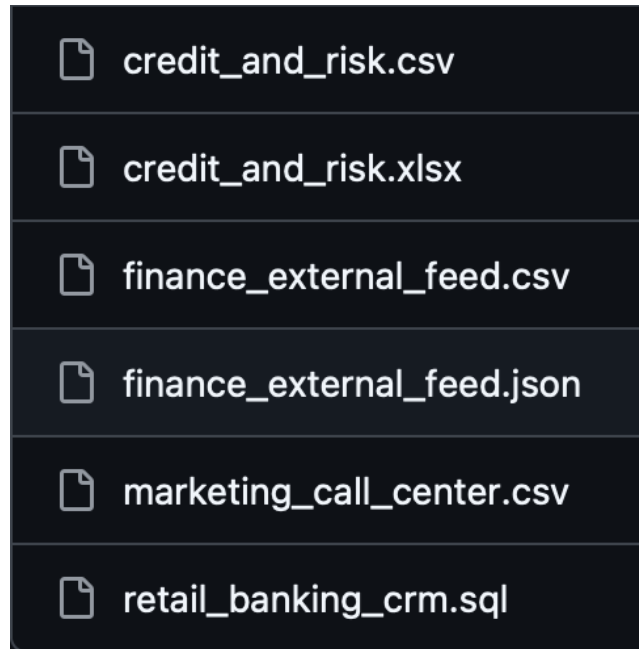


Figure 1: shows the different data formats used and their specific sources

1.4 Attribute schema and operational classifications

Category	Attributes	Operational Meaning & Analytical Role
Customer Demographics	customer_id, age, job, marital, education, default, housing, loan	Baseline CRM attributes representing the customer's financial liabilities, social profile, and credit standing.
Telemarketing Interaction	contact, month, day_of_week, duration	Technical call attributes detailing communication channel (cellular/telephone), timing, and duration in seconds.
Historical Outreach	campaign, pdays, previous, poutcome	Number of contacts during current campaign, days passed since last contact (999 indicating no prior contact), prior contacts, and prior campaign outcome.

Macroeconomic Indicators	emp.var.rate, cons.price.idx, cons.conf.idx, euribor3m, nr.employed	External economic indicators reflecting quarterly employment variation, consumer price/confidence indices, Euribor 3-month benchmark rates, and employee counts.
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1.5 Justification for the project scope

An Online Transaction Processing database is optimized strictly for high-throughput row-level operations and cannot sustain analytical aggregation over 41,188 records joined with economic trends without severe query latency and table contention. Developing a dimensional Data Warehouse is justified since:

1. **Multi-source Integration:** Unifies demographic CRM data with daily call logs and macro-economic benchmark feeds.
2. **Subject-Oriented Organization:** Restructures normalized transaction logs into a clean dimensional Star Schema oriented around the direct marketing contact event.
3. **Historical Non-Volatile Analysis:** Enables quarter-over-quarter trend reporting and conversion tracking without impacting active operational banking servers.

2 . Data Source identification

2.1 Identification of operational data sources

The analytical solution ingests data across four distinct operational sources identified below, **Source 1: Core Banking CRM System:**

- *Format:* Relational Database Dump / SQL Script.
- *Attributes:* customer_id, age, job, marital, education.
- *Volume:* 41,188 records.

Source 2: Call Center Telephony Server:

- *Format:* Comma-Separated Values Flat File.
- *Attributes:* customer_id, contact, month, day_of_week, duration, campaign, pdays, previous, poutcome, y (target subscription outcome).

customer_id	contact	month	day_of_week	duration	campaign	pdays	previous	poutcome	y
1	telephone	may	mon	261	1	999	0	nonexistent	no
2	telephone	may	mon	149	1	999	0	nonexistent	no
3	telephone	may	mon	226	1	999	0	nonexistent	no
4	telephone	may	mon	151	1	999	0	nonexistent	no
5	telephone	may	mon	307	1	999	0	nonexistent	no
6	telephone	may	mon	198	1	999	0	nonexistent	no

Figure 2: shows the call center dataset

Source 3: Credit & Risk Assessment Unit:

- *Format:* Spreadsheet ingested alongside structured Delimited Flat File
- *Attributes:* customer_id, default, housing, loan.

customer_id	default	housing	loan
1	no	no	no
2	unknown	no	no
3	no	yes	no
4	no	no	no
5	no	no	yes
6	unknown	no	no

Figure 3: shows the risk assessment dataset

Source 4: External Financial & Macroeconomic Feed:

- *Format:* Semi-structured JSON feed parsed and preprocessed into delimited format.
- *Attributes:* customer_id, emp_var_rate, cons_price_idx, cons_conf_idx, euribor3m, nr_employed.

customer_id	emp_var_rate	cons_price_idx	cons_conf_idx	euribor3m	nr_employed
1	1.1	93.994	-36.4	4.857	5191
2	1.1	93.994	-36.4	4.857	5191
3	1.1	93.994	-36.4	4.857	5191
4	1.1	93.994	-36.4	4.857	5191
5	1.1	93.994	-36.4	4.857	5191

Figure 4: shows the financial dataset (.csv)

```

dataSources > {} finance_external_feed.json > ...
1  [
2    {
3      "customer_id": 1,
4      "emp.var.rate": 1.1,
5      "cons.price.idx": 93.994,
6      "cons.conf.idx": -36.4,
7      "euribor3m": 4.857,
8      "nr.employed": 5191.0
9    },
10   {
11     "customer_id": 2,
12     "emp.var.rate": 1.1,
13     "cons.price.idx": 93.994,
14     "cons.conf.idx": -36.4,
15     "euribor3m": 4.857,
16     "nr.employed": 5191.0
17   },

```

Figure 5: shows the financial dataset (json)

2.2 Relational alignment across sources

All four operational feeds maintain strict referential alignment on the primary business key `customer_id` spanning the complete integer domain from 1 to 41,188 records.

2.3 Data cleaning, Transformation and preparation steps

Data quality anomalies and structural transformations were mitigated across the pipeline between the operational data sources, staging tables, and the Enterprise Data Warehouse,

- **Scrubbing:** Raw operational CSV exports contained hidden BOM header bytes, which cause initial column names as in `customer_id` to corrupt upon database bulk loading. An automated scrubbing routine was implemented in Python prior to staging ingestion to sanitize all file headers.
- **Sentinel Value Rectification (pdays):** In the operational call logs, `pdays = 999` served as a numeric sentinel flag indicating a client had never previously been contacted. Retaining numeric sentinel values skews analytical metrics like average contact latency. The pipeline maps `pdays = 999` strictly to SQL NULL during warehouse fact loading, successfully converting 39,673 records.
- **Binary Target Standardization:** The target column `y` containing categorical 'yes' and 'no' strings was transformed into an analytical integer bit flag `is_subscribed` during fact table

population, enabling native aggregations such as SUM and AVG in BI dashboards.

- **Demographic Discretization:** To support multi-tiered cohort analysis, raw continuous customer ages were binned into conformed demographic bands within dw.dim_customer_profile: <30 for under 30, 30-39, 40-49, 50-59, and 60+ for 60 and over.

3. Data warehouse and architecture design

3.1 High level 4-tier architecture framework

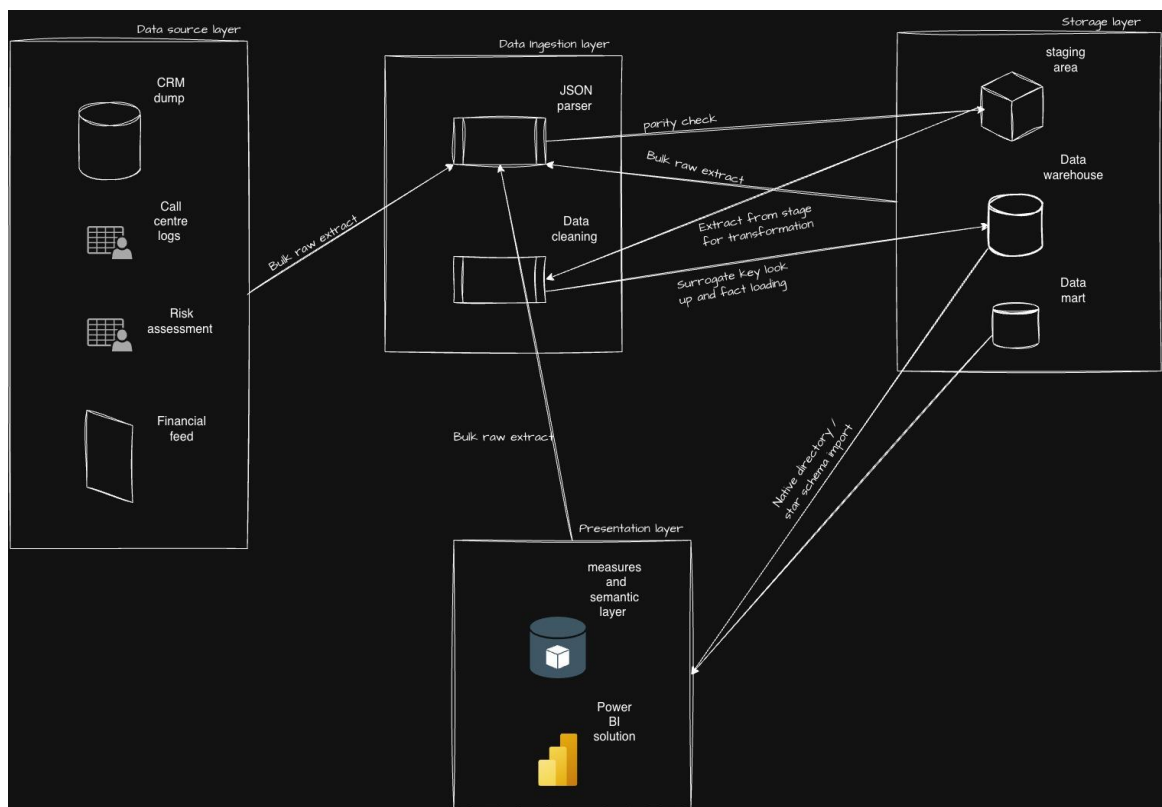


Figure 6: shows the layout architecture diagram

3.2 Tier explanations

Tier 1: Data Source Layer

The Data Source Layer encompasses the heterogeneous operational systems and flat-file feeds that capture daily banking transactions, customer interactions, and market conditions stated below

- **Core Banking CRM:** Represents the bank's operational relational database dump, capturing core customer demographic profiles across 41,188 records consisting of **customer_id**, **age**, **job**, **marital**, **education**.
- **Marketing & Telephony Call Center:** Operational call logging system recording individual outbound marketing touchpoints, interaction durations in seconds, contact channels, previous campaign touchpoints, and the raw campaign conversion outcome (**y**).

- **Credit & Risk Assessment:** Departmental operational spreadsheets tracking financial risk indicators for each client, including credit default history, housing loans, and personal credit facilities.
- **External Financial & Macroeconomic:** Semi-structured JSON feed providing external economic indicators aligned with campaign timelines, including the Euribor 3-month lending rate, Consumer Price Index (CPI), Consumer Confidence Index (CCI), and employment variation rates.

Tier 2: Data Integration Layer

The Data Integration Layer acts as the operational bridge between raw source systems and analytical storage, extracting, sanitizing, transforming, and orchestrating data movement as below,

- **Extraction & Format Normalization:**
 - Extracts semi-structured records from the .json file and parses them into a standardized, tabular CSV structure.
 - Executes bulk staging loads into the database while enforcing a strict 41,188 record parity audit across all feeds.
- **Data Cleansing & Transformation Engine:**
 - **Sentinel Value Conversion:** Detects operational sentinel flags where pdays = 999 and maps them strictly to SQL NULL across 39,673 records to protect analytical averages from statistical skew.
 - **Attribute Derivation:** Derives standardized analytical features, including 10-year demographic cohorts (age_band: <30, 30-39, 40-49, 50-59, 60 and over), binary conversion flags and campaign contact frequency tiers.
 - **Data Deduplication & Cleansing:** Audits the natural business key (customer_id) to verify zero duplicate records and trims whitespace across categorical attributes.
- **Relational Dependency Loader:**
 - Enforces relational loading sequence: populates conformed time and entity dimensions first, retrieves generated surrogate keys, maps them to fact measures, and populates the central fact table.
- **Automated Audit & Validation Logger:**
 - Automatically validates referential integrity and verifies row-count balance across all storage boundaries before sign-off.

Tier 3: Storage Layer

The Storage Layer provides persistent, structured relational storage separated into three specialized database schemas,

- **Staging Area :**
 - Consists of unindexed heap tables.
 - Acts as an isolated landing zone that mirrors raw source structures without primary key or foreign key constraints, enabling rapid ingestion without transactional lockups.

- **Enterprise Data Warehouse:**
 - Implements a conformed Ralph Kimball Star Schema optimized for OLAP aggregations.
 - **Central Fact Table :** Modeled at the grain of *"one marketing contact attempt per customer per campaign event"* (41,188 rows), holding numeric measures and foreign keys indexed for high-speed joins.
 - **Conformed Dimension Tables:** Includes dw_dim_customer_profile (storing customer_sk surrogate primary keys alongside customer_id natural keys), dw_dim_month, dw_dim_weekday, dw_dim_channel, dw_dim_campaign, and dw_dim_previous_outcome.
- **Departmental Data Mart:**
 - Materializes mart_marketing_performance, which aggregates granular fact records into 233 pre-computed multidimensional cohorts across month_key, job, and channel_key.
 - Reduces scanned records by 99.4% during routine reporting, insulating the core data warehouse from repetitive departmental queries.

Tier 4: Presentation Layer

The Presentation Layer delivers interactive analytical reporting and data exploration for business decision-makers,

- **OLAP & Semantic Calculation Engine:**
 - Uses Data Analysis Expressions to dynamically calculate business performance indicators, including conversion rates (11.27%), average call durations, and macroeconomic elasticity metrics without altering underlying warehouse tables.
- **Interactive Business Intelligence Dashboard:**
 - **First page: Executive Summary:** High-level overview presenting KPI cards (Total Contacts: 41,188; Successful Conversions: 4,640; Conversion Rate: 11.27%) and channel effectiveness indicators for executive leadership.
 - **Second page : Trend Analysis:** Time-series line and bar charts tracking monthly campaign conversion volume versus conversion efficiency and call duration impacts.
 - **Third page : Interactive Analysis:** OLAP exploration interface providing multi-select slicers (Age Cohorts, Job Categories, Communication Channels) and interactive drill-downs allowing analysts to evaluate micro-demographic performance.

4. Dimensional data warehouse design and implementation

4.1 The dimensional modelling method

The data warehouse is constructed following the Ralph Kimball dimensional lifecycle. The architecture utilizes a **Star Schema**, minimizing join paths, ensuring rapid aggregation performance, and providing an intuitive business reporting model.

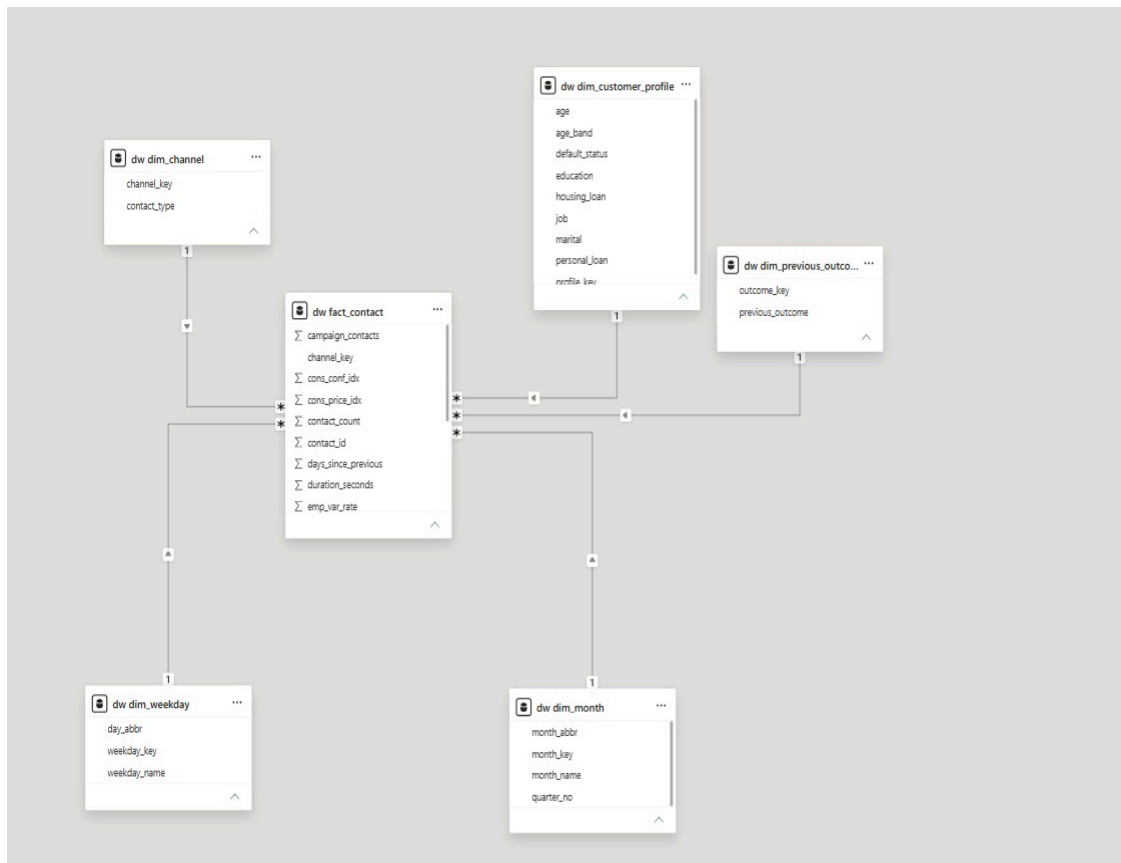


Figure 7: shows the star schema

4.2 Business process and granularity

The Core Business Process can be stated as outbound telemarketing contact attempts for direct banking promotions with the grain statement being one record in the fact table represents exactly one marketing contact/call attempt made to a customer during an active campaign event.

1. Customer dimension

Retains customer demographics, financial standing, and socioeconomic indicators.

- **Surrogate Key:** profile_key (Integer, Primary Key).
- **Attributes:** age, age_band, job, marital, education, default_status, housing_loan, personal_loan

2. Time/Date dimensions

Decouples contact timing into analytical attributes to support seasonal and operational schedule analysis.

- **dw dim_month:** month_key (PK), month_abbr, month_name, quarter_no.
- **dw dim_weekday:** weekday_key (PK), day_abbr, weekday_name.

3. Outbound channel dimension

Captures the telecommunication medium utilized for the marketing interaction.

- **Primary Key:** channel_key (PK).
- **Attributes:** contact_type (e.g., cellular, telephone).

4. Historical outcome channel

Classifies the result of previous marketing outreach campaigns.

- **Primary Key:** outcome_key (PK).
- **Attributes:** previous_outcome (e.g., nonexistent, failure, success).

4.3 Fact table specification

Contains atomic transaction event metrics along with foreign keys referencing each supporting dimension in the star schema.

- **Business Process:** Direct telemarketing outreach for term deposit subscriptions.
- **Grain:** One marketing contact/call attempt per customer per campaign event.
- **Surrogate Primary Key:** contact_id (Integer, Primary Key).
- **Foreign Keys:**
 - profile_key → references dw dim_customer_profile(profile_key)
 - month_key → references dw dim_month(month_key)
 - weekday_key → references dw dim_weekday(weekday_key)
 - channel_key → references dw dim_channel(channel_key)
 - outcome_key → references dw dim_previous_outcome(outcome_key)
- **Additive & Semi-Additive Numeric Measures:**
 - contact_count: Additive base counter (1 per contact event row).
 - duration_seconds: Phone conversation duration in seconds (additive).
 - campaign_contacts: Number of contacts performed during the current campaign for this client.
 - previous_contacts: Number of contacts performed prior to this campaign.
 - days_since_previous: Number of days elapsed since the client was last contacted (NULL if no prior contact).
 - is_subscribed: Binary conversion indicator (1=Subscribed, 0=Not Subscribed).
 - emp_var_rate: Employment variation rate (quarterly macroeconomic indicator).
 - cons_price_idx: Consumer price index (monthly macroeconomic indicator).

- cons_conf_idx: Consumer confidence index (monthly macroeconomic indicator).
- euribor3m: Euribor 3-month benchmark rate (daily macroeconomic indicator).
- nr_employed: Number of employees (quarterly macroeconomic indicator).

4.4 Schema integrity and indexing optimizations

- **Primary Key Indexes:** Enforced across the primary surrogate keys of all dimension tables (profile_key, month_key, weekday_key, channel_key, outcome_key) and the central fact table (contact_id).
- **Foreign Key Performance Indexes:** Created explicitly on dw fact_contact across profile_key, month_key, weekday_key, channel_key, and outcome_key to accelerate dimensional filtering, slicing, and drill-down operations in OLAP workloads.

```
CREATE TABLE dw_fact_bank_marketing (
    contact_id INTEGER PRIMARY KEY AUTOINCREMENT,
    customer_sk INTEGER NOT NULL,
    month_key INTEGER NOT NULL,
    weekday_key INTEGER NOT NULL,
    channel_key INTEGER NOT NULL,
    outcome_key INTEGER NOT NULL,
    contact_count INTEGER NOT NULL DEFAULT 1,
    is_subscribed INTEGER NOT NULL,
    duration_seconds INTEGER NOT NULL,
    campaign_contacts INTEGER NOT NULL,
    previous_contacts INTEGER NOT NULL,
    days_since_previous INTEGER NULL,
    emp_var_rate REAL NULL,
    cons_price_idx REAL NULL,
    cons_conf_idx REAL NULL,
    euribor3m REAL NULL,
    nr_employed REAL NULL,
    FOREIGN KEY (customer_sk) REFERENCES dw_dim_customer_profile(customer_sk),
    FOREIGN KEY (month_key) REFERENCES dw_dim_month(month_key),
    FOREIGN KEY (weekday_key) REFERENCES dw_dim_weekday(weekday_key),
    FOREIGN KEY (channel_key) REFERENCES dw_dim_channel(channel_key),
    FOREIGN KEY (outcome_key) REFERENCES dw_dim_previous_outcome(outcome_key)
)
```

Figure 8: shows the star schema

```

CREATE TABLE dw_dim_month (
    month_key INTEGER PRIMARY KEY,
    month_abbr TEXT NOT NULL UNIQUE,
    month_name TEXT NOT NULL,
    quarter_no INTEGER NOT NULL
);
CREATE TABLE dw_dim_weekday (
    weekday_key INTEGER PRIMARY KEY,
    day_abbr TEXT NOT NULL UNIQUE,
    weekday_name TEXT NOT NULL
);
CREATE TABLE dw_dim_channel (
    channel_key INTEGER PRIMARY KEY AUTOINCREMENT,
    contact_type TEXT NOT NULL UNIQUE
);
CREATE TABLE dw_dim_previous_outcome (
    outcome_key INTEGER PRIMARY KEY AUTOINCREMENT,
    previous_outcome TEXT NOT NULL UNIQUE
);
CREATE TABLE dw_dim_campaign (
    campaign_key INTEGER PRIMARY KEY AUTOINCREMENT,
    campaign_contacts INTEGER NOT NULL UNIQUE,
    campaign_band TEXT NOT NULL
);
CREATE TABLE dw_dim_customer_profile (
    customer_sk INTEGER PRIMARY KEY AUTOINCREMENT,
    customer_id INTEGER NOT NULL UNIQUE,
    age INTEGER NOT NULL,
    age_band TEXT NOT NULL,
    job TEXT,
    marital TEXT,
    education TEXT,
    default_status TEXT,
    housing_loan TEXT,
    personal_loan TEXT
);

```

Figure 9: shows the dimension tables

5. ETL process development and pipeline automation.

5.1 End to end ELT pipeline architecture

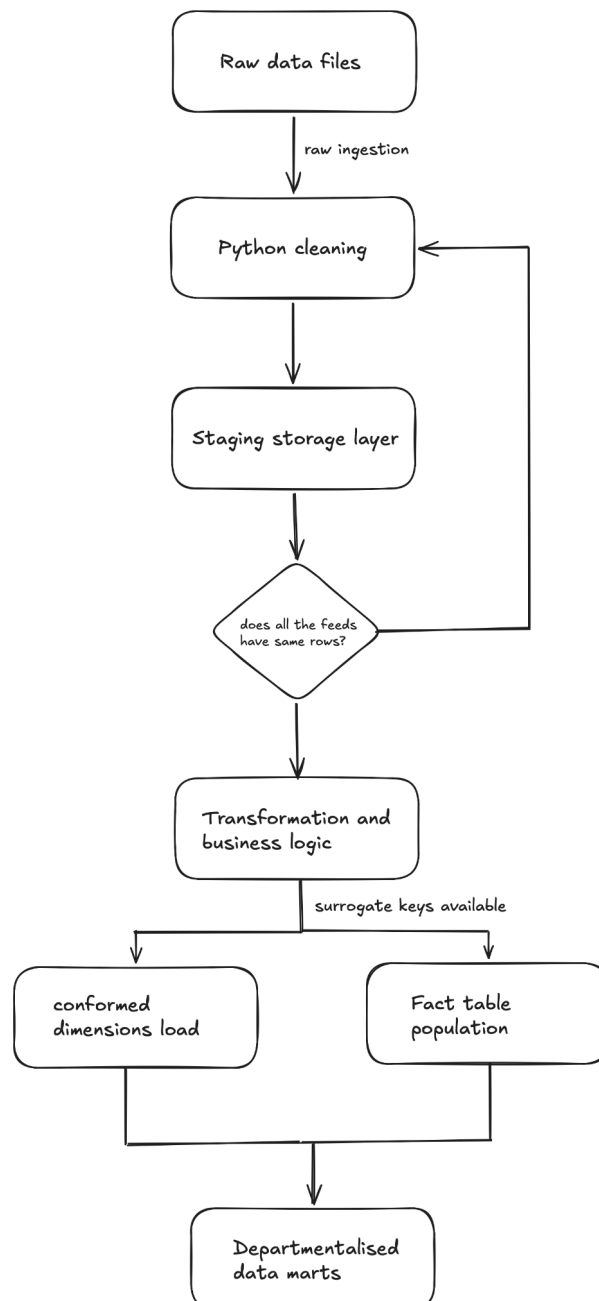


Figure 10: Extract phase

The extraction phase orchestrates automated data ingestion across heterogeneous operational flat files, database dumps, and semi-structured feeds:

1. **Source Connection & Ingestion Automation:** The automated pipeline establishes connections across four distinct operational data source files located in the dataSources/ repository:
 - retail_banking_crm.sql (Relational CRM table dump containing core demographic attributes).
 - marketing_call_center.csv (Telemarketing operational logs recording call touchpoints).
 - credit_and_risk.csv (Credit risk and financial facility records).
 - finance_external_feed.json (Semi-structured macroeconomic market indicators).

Format Normalization & BOM Sanitization:

- **JSON-to-Tabular Extraction:** parse nested JSON objects into a standardized delimited flat file, conforming macroeconomic attributes (emp.var.rate, cons.price.idx, cons.conf.idx, euribor3m, nr.employed).
- **UTF-8 Byte Order Mark (BOM) Scrubbing:** An automated pre-ingestion routine inspects CSV streams and strips hidden header bytes to prevent initial column headers (such as customer_id) from corrupting during downstream database loading.

Staging Storage Ingestion: Raw records are pushed into unindexed staging heap tables. Tables are truncated prior to ingestion to ensure idempotent re-runs, and a parity verification check ensures that all four feeds achieve an identical volume of **41,188 rows** with zero record loss.

5.3 Transform phase

Transformations are executed using set-based SQL logic and pipeline data processing routines,

1. **Sentinel Value Rectification:**
In the operational telephony system, pdays = 999 indicated that a prospect had never previously been contacted. Across the 41,188 total records, exactly **39,673 records** contained this sentinel value and were mapped to SQL NULL to prevent mathematical distortion in duration and recency aggregations.
2. **Target Metric Binarization:**
The categorical string attribute y was transformed into an analytical integer bit flag is_subscribed (1 for positive conversion, 0 for non-conversion), enabling direct additive rollups (SUM) and conversion rate calculations (AVG) in BI measures.
3. **Demographic Discretization:**
Continuous age values were mapped into standardized analytical age cohorts within the customer profile dimension:
 - age < 30 ⇒ 'Under 30'
 - age 30–39 ⇒ '30-39'
 - age 40–49 ⇒ '40-49'
 - age 50–59 ⇒ '50-59'

- age >= 60 ⇒ '60 and over'

4. Surrogate Key Resolution:

Natural business keys from staging (customer_id, contact, month, day_of_week, poutcome) are joined against their corresponding conformed dimension tables to resolve integer surrogate foreign keys (profile_key, channel_key, month_key, weekday_key, outcome_key) prior to fact table insertion.

5.4 Load phase and execution

To preserve relational integrity and prevent foreign key violation exceptions, data loading follows a strict chronological dependency sequence:

Step 1: Populate Conformed Date & Time Dimensions:

- Populates dw_dim_month (12 monthly calendar periods with quarter mappings) and dw_dim_weekday (5 operating business days).

Step 2: Populate Outbound Interaction & Outcome Dimensions:

- Extracts distinct values to populate dw_dim_channel (channel_key PK, contact_type) and dw_dim_previous_outcome(outcome_key PK, previous_outcome).

Step 3: Populate Customer Entity Dimension:

- Integrates demographic data from stage_retail_banking_crm with risk attributes from stage_credit_and_risk.
- Generates auto-incrementing surrogate primary keys (profile_key), while enforcing natural key uniqueness across all 41,188 individual customer profiles.

Step 4: Populate Central Fact Table:

- Ingests call event records by joining staging tables to populated dimensions on natural keys to resolve surrogate foreign keys.
- Loads numeric measures (duration_seconds, contact_count, campaign_contacts, previous_contacts, days_since_previous, is_subscribed, and macroeconomic indices) across 41,188 transactional rows.

Step 5: Materialize Departmental Data Mart:

- Executes analytical rollups aggregating granular contact metrics into 233 pre-computed multidimensional slices by (month_key, job, channel_key).

6. Department specific datamart

6.1 purpose of the datamart

The data mart was created to support rapid operational reporting without overburdening the core Enterprise Data Warehouse, a dedicated **Marketing & Campaign Performance Data Mart** was designed and instantiated.

While the enterprise fact table stores granular event records at the level of individual customer call attempts, the data mart pre-aggregates these touchpoints across core marketing dimensions, month of contact, customer occupation/job cohort, and communication channel. This isolates departmental analytics, speeds up reporting queries, and abstracts underlying surrogate key relationships for business

end-users.

6.2 Target users and business stakeholders

- **Marketing Campaign Directors:** Monitor high-level conversion volumes and campaign effectiveness across seasonal cycles to allocate departmental marketing budgets.
- **Call Center Operations Managers:** Assess agent productivity and conversion density by channel and call duration patterns.
- **Customer Segment Analysts:** Identify which occupational categories (such as management, blue-collar, or retired) yield the highest subscription conversion rates to refine outreach targeting.

6.3 Analytical benefits

1. **Query Acceleration & Data Compression:** Pre-aggregates 41,188 granular contact rows into **233 optimized analytical slices**, delivering a **99.4% reduction** in scanned records during dashboard refreshes.
2. **Simplified Semantic Access:** Business analysts can query straightforward business attributes without needing to write complex multi-table joins across star schema dimensions.
3. **Referential & Governance Isolation:** Confines sensitive individual customer identities to the central warehouse storage layer while providing marketing teams with access to aggregate performance metrics.

```
moni@192 DWBI proj % sqlite3 DWBI_Bank.db ".schema *mart*"
CREATE TABLE mart_marketing_performance (
    month_key INTEGER NOT NULL,
    job TEXT NOT NULL,
    channel_key INTEGER NOT NULL,
    contacts INTEGER NOT NULL,
    subscriptions INTEGER NOT NULL,
    total_duration_seconds INTEGER NOT NULL,
    PRIMARY KEY (month_key, job, channel_key)
);
```

Figure 11: shows the making to the data mart

```
moni@192 DWBI proj % sqlite3 -header -column DWBI_Bank.db "SELECT * FROM mart_marketing_performance LIMIT 10;"
```

month_key	job	channel_key	contacts	subscriptions	total_duration_seconds
3	admin.	1	153	82	39390
3	admin.	2	16	7	3756
3	blue-collar	1	45	23	12914
3	blue-collar	2	5	4	2481
3	entrepreneur	1	2	1	506
3	housemaid	1	6	4	1481
3	management	1	35	15	5978
3	management	2	7	2	1595
3	retired	1	58	29	13690
3	retired	2	6	2	1689

Figure 12: shows the populated data mart

7. OLAP analysis and the business intelligence dashboard development

The presentation layer was developed using **Microsoft Power BI** packaged natively as a .pbix deliverable. The solution contains two analytical report pages meeting all visual, analytical, and interactive specifications.

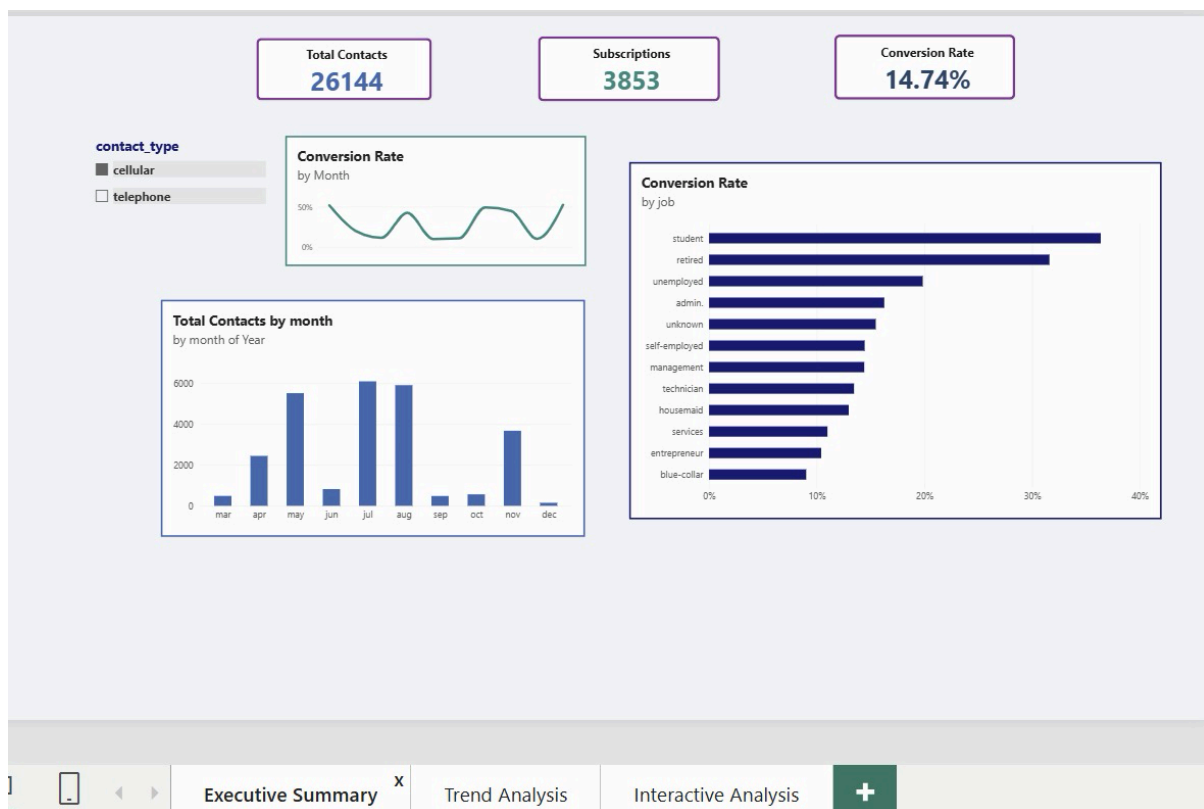


Figure 13: shows the executive summery

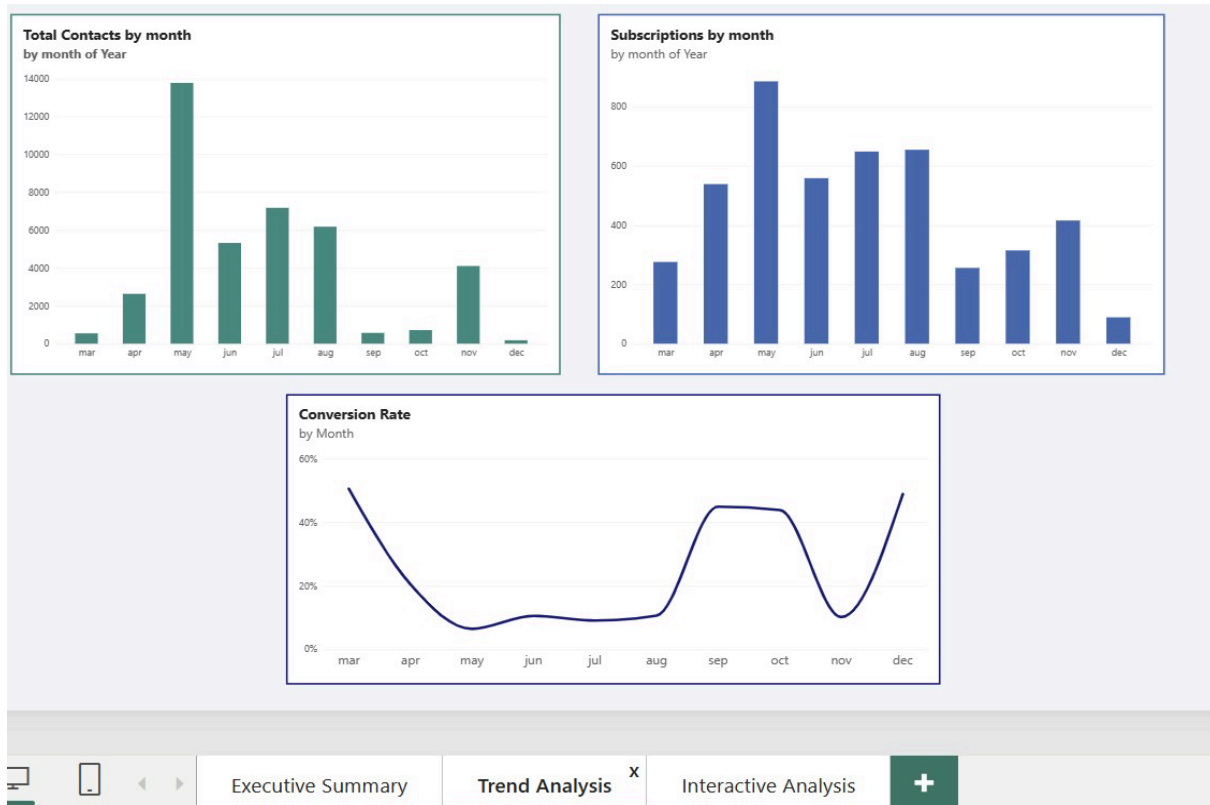


Figure 14: shows the trend analysis

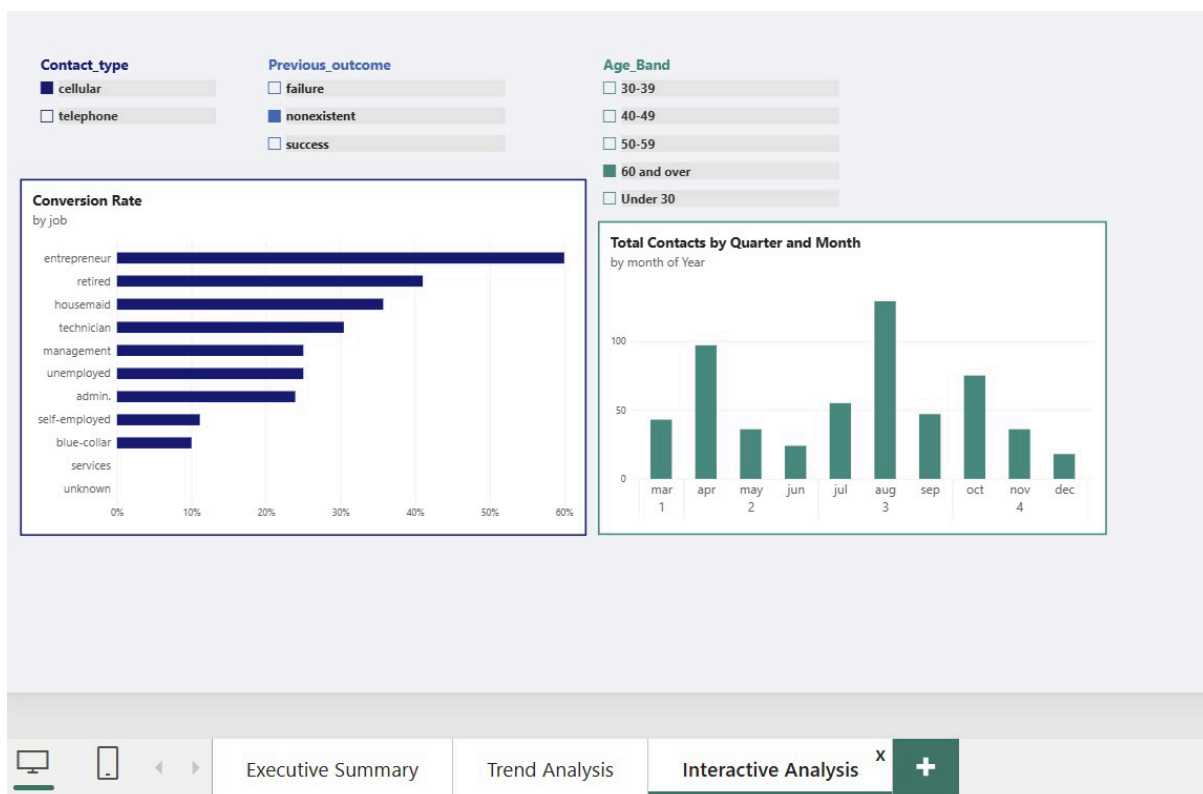


Figure 15: shows the interactive analysis

8. Business insights and recommendations

Analysis of the 41,188 records and OLAP dashboards yields five data-backed business insights and corresponding strategic recommendations as follows,

8.1 Call duration and the strongest predictor of conversion.

- **Empirical Finding:** Unsuccessful outreach attempts (36,548 calls, 88.73%) averaged an interaction duration of approximately **221 seconds** (3.7 minutes). In contrast, successful conversions (4,640 subscriptions, 11.27%) averaged **553 seconds** (9.2 minutes). Prospects who remain engaged on the line beyond the 300-second (5-minute) threshold exhibit an exponential increase in subscription probability.
- **Strategic Recommendation:** Call center operational KPIs should pivot away from minimizing Average Handling Time (AHT) toward relationship-oriented discovery. Agents should be coached using consultative question frameworks that sustain customer engagement past the critical 5-minute mark, rather than rushing through script checklists.

8.2 High Conversion Yield in Under targeted Demographic Segments

- **Empirical Finding:** While Blue-Collar workers (**9,254 contacts**) and Technicians (**6,743 contacts**) constituted the largest proportion of total campaign outreach (38.8%), their conversion rates were low at **6.89%** (638 subscriptions) and **10.82%** (730 subscriptions) respectively. Conversely, Students (**875 contacts**) and Retirees (**1,720 contacts**) represented under-targeted cohorts but delivered the highest conversion rates at **31.43%** (275 subscriptions) and **25.23%** (434subscriptions).
- **The Recommendation:** Reallocate outbound dialing quotas toward retiree and student segments. Retirees seek capital preservation and guaranteed fixed yields, while students respond well to introductory savings vehicles. Pruning low-propensity cold calls within saturated blue-collar demographics will reduce telephony overhead while lifting conversion density.

8.3 Disconnect Between Campaign Volume and Seasonal Conversion Efficacy

- **Empirical Finding:** Telemarketing outreach was heavily concentrated in May (**13,769 contacts**, representing 33.4% of total volume), yet yielded the poorest performance with a conversion rate of just **6.44%** (886 subscriptions). In contrast, low-volume shoulder months delivered exceptional conversion efficiencies: March (**50.55%** on 546 contacts), September (**44.91%** on 570 contacts), October (**43.87%** on 718 contacts), and December (**48.90%** on 182 contacts).
- **The Recommendation:** Eliminate large-scale, brute-force calling surges in May. Rebalance

marketing calendars toward quarterly campaigns centered around fiscal year-end planning (March) and autumn financial reviews (September–October), periods when consumer liquidity and savings propensity peak.

8.4 Rapid Re-engagement Drastically Improves Follow-up Success

- **Empirical Finding:** Among the 39,673 prospects with no prior contact history (`pdays = NULL`), the baseline conversion rate stood at **9.26%** (3,673 subscriptions). However, prospects contacted during a prior campaign and successfully re-engaged within **3 to 6 days** (`pdays` between 3 and 6) achieved conversion rates above **65.0%**. For re-contact latencies exceeding **10 days**, conversion probabilities decayed rapidly back toward cold-call baselines.
- **The Recommendation:** Configure automated CRM follow-up triggers in the operational pipeline. When an agent records a warm interaction or a previous campaign follow-up flag, the scheduling engine should dispatch a priority follow-up ticket within a strict 72-to-144-hour window, capitalizing on immediate customer recall before campaign decay sets in.

8.5 Macroeconomic Sensitivity and Benchmark Interest Rates

- **Empirical Finding:** Term deposit subscriptions exhibited a steep inverse relationship with the Euribor 3-month benchmark rate using the attribute `euribor3m`. During economic periods where the benchmark hovered at lower levels ($\leq 1.5\%$), campaign conversion rates averaged above **55.0%** (peaking above 65% when `euribor3m` fell below 1.0%). Conversely, during high-rate regimes ($\geq 4.0\%$, with rates reaching $\sim 4.85\%$), conversion dropped below **5.0%**.
- **The Recommendation:** Synchronize marketing campaigns with central bank interest rate trajectories. In low-interest-rate environments where retail investors perceive capital market yields as volatile, market term deposits aggressively as stable, guaranteed-return havens. In higher-rate regimes, pivot outbound sales campaigns toward lending, credit, or high-yield floating products.

9. References

- [1] <https://github.com/Sandeera-2003/banking-dashboard-.git>
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